In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

5.1. Starting from the matrix form of the projections and reflections described in this chapter, deduce the matrices of

a) the orthogonal projections on the coordinate axes and on the coordinate hyperplanes of $\mathcal{K}$.

b) the orthogonal reflections in the coordinate axes and in the coordinate hyperplanes of $\mathcal{K}$.

5.2. Consider the vector $\mathbf{v}(2,1,1)$.

a) Give the matrix form for the parallel projection on the plane $\pi : z = 1$ parallel to $\mathbf{v}$.

b) Give the matrix form for the parallel reflection in the plane $\pi : z = 0$ parallel to $\mathbf{v}$.

5.3. Determine the orthogonal projection of the line $\ell : 2x - y - 1 = 0 \cap x + y - z + 1 = 0$ on the plane $\pi : x + 2y - z = 0$. Do this by determining the matrix form of the projection and discuss all other options.

5.4. Let H be a hyperplane and let $\mathbf{v}$ be a vector which is not parallel to H. Use the deduced matrix forms to show that

a) $\mathrm{Pr}_{H,\mathbf{v}} \circ \mathrm{Pr}_{H,\mathbf{v}} = \mathrm{Pr}_{H,\mathbf{v}}$ and

b) $\mathrm{Ref}_{H,\mathbf{v}} \circ \mathrm{Ref}_{H,\mathbf{v}} = \mathrm{Id}$.

5.5. Let $\phi(\mathbf{x}) = A\mathbf{x} + b$ be an affine transformation. Give the homogeneous matrix of the inverse transformation ϕ^{-1}.

5.6. Let π be a plane in $\mathbb{E}^3$ with normal vector $\mathbf{n}$ and equation $\pi : \langle \mathbf{n}, \mathbf{x} \rangle = c$. Show that the composition of the orthogonal reflection in π followed by a translation by a vector $\mathbf{v} \parallel \mathbf{n}$ is a reflection in a plane π' and deduce an equation of π'.

5.7. Let H be a hyperplane passing through the origin. Let $\mathbf{n}$ be a normal vector for H of length 1. Starting from the matrix forms of projections and reflections, deduce that

$$\mathrm{Pr}^{\perp}_{H}(P) = \big(I_n - \mathbf{n} \otimes \mathbf{n}\big)P \quad \text{and} \quad \mathrm{Ref}^{\perp}_{H}(P) = \big(I_n - 2 \cdot \mathbf{n} \otimes \mathbf{n}\big)P.$$